

Redmi 4X Level 3 Maintenance

Guide V01

Technical support internal document control: TSIMHNA13 Redmi
4X Level 3 Maintenance Guidance V01 Scope of application:
Change history of analysis center,
motherboard and whole machine
maintenance factory:

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Redmi 4X Level 3 Maintenance Guide V01

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1. Introduction of basic information

1.1 Product Overview

Product Overview:

参数类	参数项	红米4X(2GB+16GB)	红米4X(3GB+32GB)
外观	外观		
	屏幕尺寸	5.0英寸	5.0英寸
	外观尺寸	139.24 mm*69.96mm*8.65mm	139.24 mm*69.96mm*8.65mm
	重量	150g	150g
	后壳材质	三段式 (上下塑料, 中段金属)	三段式 (上下塑料, 中段金属)
	屏幕类型	IPS	IPS
	屏幕保护玻璃类型	2.5D保护玻璃	2.5D保护玻璃
	贴合类型	全贴合 (GFF)	全贴合 (GFF)
	背光技术	支持 (基础版)	支持 (基础版)
	护眼模式	支持	支持
	PPI	296	296
	屏幕分辨率	1280*720	1280*720
网络制式	色彩	1600万色	1600万色
	触摸屏	电容式多点触控屏	电容式多点触控屏
	网络制式支持	TDD-LTE/FDD-LTE/TD-SCDMA/ WCDMA/CDMA2000/GSM/CDMA1X	TDD-LTE/FDD-LTE/TD-SCDMA/ WCDMA/CDMA2000/GSM/CDMA1X
	网络频率	TDD-LTE: B38/B39/B40/B41(2555-2655MHz) FDD-LTE: B1/B3/B5/B7/B8 TD-SCDMA: B34/B39 WCDMA: B1/B2/B5/B8 CDMA2000: 1X GSM/CDMA 1X: B2/B3/B5/B8/BC0	TDD-LTE: B38/B39/B40/B41(2555-2655MHz) FDD-LTE: B1/B3/B5/B7/B8 TD-SCDMA: B34/B39 WCDMA: B1/B2/B5/B8 CDMA2000: 1X GSM/CDMA 1X: B2/B3/B5/B8/BC0
	卡槽制式	卡槽一: TDD-LTE/FDD-LTE/TD-SCDMA/WCDMA/CDMA2000/GSM/CDMA1X 卡槽二: TDD-LTE/FDD-LTE/TD-SCDMA/WCDMA/CDMA2000/GSM/CDMA1X	卡槽一: TDD-LTE/FDD-LTE/TD-SCDMA/WCDMA/CDMA2000/GSM/CDMA1X 卡槽二: TDD-LTE/FDD-LTE/TD-SCDMA/WCDMA/CDMA2000/GSM/CDMA1X
	VoLTE	支持	支持
	全网通版本	全网通3.0 (高通)	全网通3.0 (高通)
	4G+	支持 (带内网双待机) / 双制式聚合Cat6	支持 (带内网双待机) / 双制式聚合Cat6
	网络模式	双卡双待单通	双卡双待单通
	SIM卡类型	Micro SIM卡+Nano SIM卡/Micro SD卡 (二选一)	Micro SIM卡+Nano SIM卡/Micro SD卡 (二选一)
	CPU厂家	高通	高通
SoC	SoC型号	骁龙435 MSM8940	骁龙435 MSM8940
	CPU核心数量	8核	8核
	CPU主频	最高1.4GHz	最高1.4GHz
	CPU制程工艺	28nm	28nm
	核心类型	Cortex-A53	Cortex-A53
	图形处理器	Adreno™ 505	Adreno™ 505
	RAM/内存	2GB	3GB
	内存类型	LPDDR3	LPDDR3
	内存容量	16GB	32GB
	eMMC版本	S3	S3
	扩展存储	支持最高128GB (Vfat32文件系统格式)	支持最高128GB (Vfat32文件系统格式)
	扩展卡类型	Micro SD卡 (TF卡)	Micro SD卡 (TF卡)
前置摄像头	CMOS供应商	OV/三星	OV/三星
	像素	1300万像素	1300万像素
	光圈	F2.0	F2.0
	对焦方式	相位对焦+反差对焦	相位对焦+反差对焦
	滤镜	支持	支持
	慢速支持	1080P	1080P
	闪光灯	LED补光灯	LED补光灯
	全景模式	支持	支持
	水平仪	支持	支持
	HDR模式	支持	支持
	美颜模式	支持	支持
	倒计时拍摄	支持	支持
后置摄像头	手动白平衡	支持	支持
	手动感光度	支持	支持
	实时滤镜	支持	支持
	智能美颜	36级实时智能美颜	36级实时智能美颜
	像素	500万像素	500万像素
	光圈	F2.2	F2.2
	慢速支持	1080P	1080P
	智能美颜	36级实时智能美颜	36级实时智能美颜
	WiFi 2.4GHz	802.11b/g/n	802.11b/g/n
	WiFi 5GHz	不支持	不支持
	NFC	不支持	不支持
	蓝牙	BT4.2+HID	BT4.2+HID
电池	电池容量	额定容量: 4000mAh 典型容量: 4100mAh	额定容量: 4000mAh 典型容量: 4100mAh
	快充技术	5V/2A快充	5V/2A快充
	陀螺仪	支持	支持
	重力感应器	支持	支持
	距离感应器	支持	支持
	光线感应器	支持	支持
	指纹识别器	支持	支持
	霍尔感应器	不支持	不支持
	气压感应器	不支持	不支持
	红外发射器	支持	支持
	指纹识别	背部按压式指纹识别	背部按压式指纹识别
	指纹录入	12次采样录入 (最多存入5个)	12次采样录入 (最多存入5个)
数据接口	USB接口类型	Micro USB 2.0 (Micro B型接口)	Micro USB 2.0 (Micro B型接口)
	OTG	支持	支持
	MHL	不支持	不支持
	卫星定位	GPS (A-GPS) + GLONASS + Beidou	GPS (A-GPS) + GLONASS + Beidou
	网络定位	支持	支持
	陀螺	双激光透透陀螺	双激光透透陀螺
	陀螺角	不支持	不支持
	可拆卸后盖	不支持	不支持
	挂绳孔	不支持	不支持
	呼吸灯	支持	支持
	FM	支持	支持
	模拟模式	支持	支持
其它	操作系统	MIUI8 (基于Android6.0)	MIUI8 (基于Android6.0)

Instructions for using dual cards:

Mobile cards can be used with Unicom cards and telecom cards at will. Any card can choose 4G network, and the other card is 2G network. Two telecom cards cannot be used at the same time: when used at the same time, one card can use 4G network, and the other card cannot register for network.

1.2 Special welding fixture for red rice 4X

Material code: SCNC030022700



1.3 Redmi 4X power supply adapter

Redmi 4X False Wire and HM Note 3 Universal.

Material code: SCNC010013700 power supply box + SCNC020010500 power supply adapter



1.4 Pasting position and specification of maintenance label

Redmi 4X
Maintenance Label:
The pasting position



1.5 Matters needing attention in motherboard maintenance

1. The first boot time of HM4X motherboard will exceed 5min after brushing. Please wait patiently and do not cut off power.
2. Before soldering the motherboard, the heat dissipation silica gel needs to be removed to prevent the chip under the heat dissipation silica gel pad from being squeezed during heating, resulting in motherboard failure.

3. Take anti-welding measures when welding components near key interfaces, display interfaces and earphone interfaces. These

Plastic components are easy to weld.

4. Redmi 4X U1101 and U1801 binding relationship:

One U1101 can correspond to multiple U1801 (unbound), and one U1801 (bound) can only correspond to one U1101. U1801 can be replaced separately when it is damaged during maintenance, while replacing U1101 when it is damaged needs to replace U1801 synchronously.

Redmi 4X binding relationship is generated in the generation stage of TEEK file (number change step). After we replace the new EMMC, as long as TEEK file is not generated, there is no binding relationship between U1101 and U1801. At this time, if U1101 fails, it can be replaced separately.

5. After the good motherboard brushes the user program, it shows that the encryption failure can be solved by re-brushing the NV factory package and re-brushing the user package after calibration.

1.6 Brush mode

Brush platform: MiFlash

HM4X has BL lock, so the mobile phone can only be brushed deeply, not Fastboot. After brushing the factory software deeply, the BL lock can be erased, and the BL lock of the user software will be generated. Fastboot Brush Error "No Error Specified Mistake".

Special factory package for renaming (keep NV package)

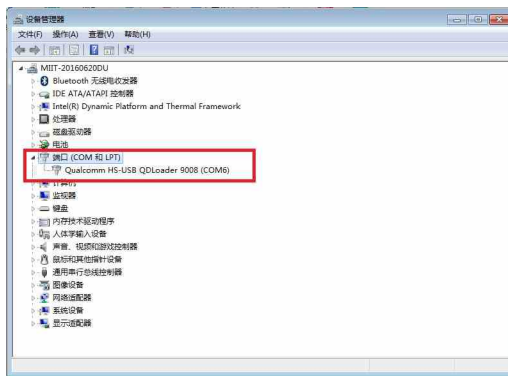
SW_S88536AA1_V013_M11_MP_XM-MIUI-FACTORY13_NV

Special Factory Package for Maintenance (Clear NV Package)

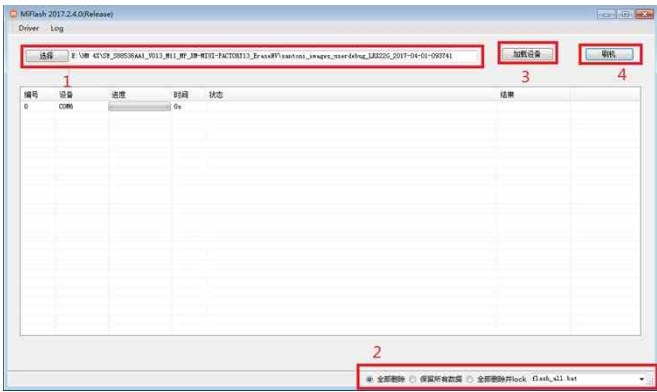
SW_S88536AA1_V013_M11_MP_XM-MIUI-FACTORY13_EraseNV

Brushing method: deep brushing

1. When the mobile phone is turned off, short-circuit TP7 and TP9, and the mobile phone enters deep brush mode (identify port 9008)



After brushing the factory package, check all delete options at the bottom of the software and click Brush Machine.

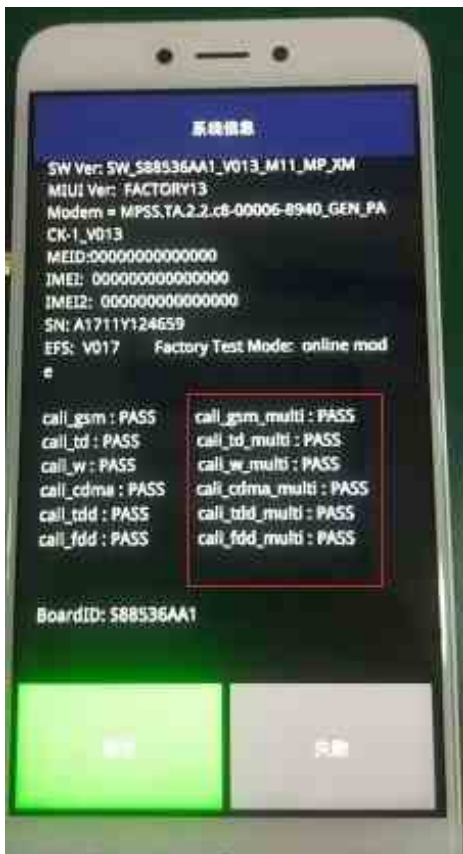


1.7 RF calibration test correlation

1. Deep Brush Factory Software (SW_S88536AA1_V013_M11_MP_XM-MIUI-FACTORY13_EraseNV), written to the FSN using the DT tool.
2. Open the calibration software, and connect the main antenna and auxiliary receiving two RF lines when the mobile phone is turned on. Connect the USB cable and start the RF calibration of the mobile phone. After RF calibration is completed, this flag bit displays PASS.



to write the flag bit of comprehensive test item, open HM2 write for A13 project. Flag bits are written in the same way as Redmi 3S. Enter the system information to see if the write was successful.

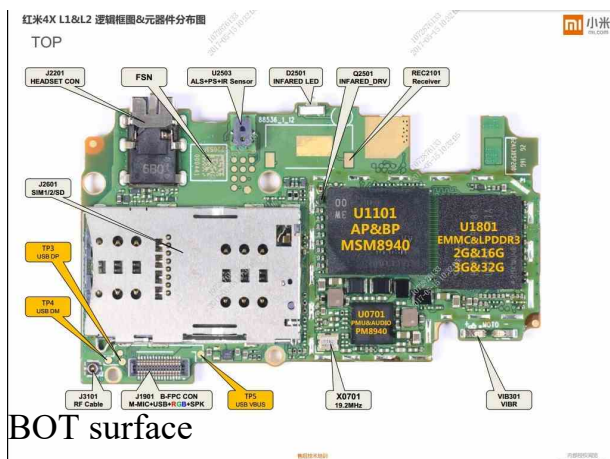


4. After calibration, to test the signal function, you need to use DT tool to restart the mobile phone.

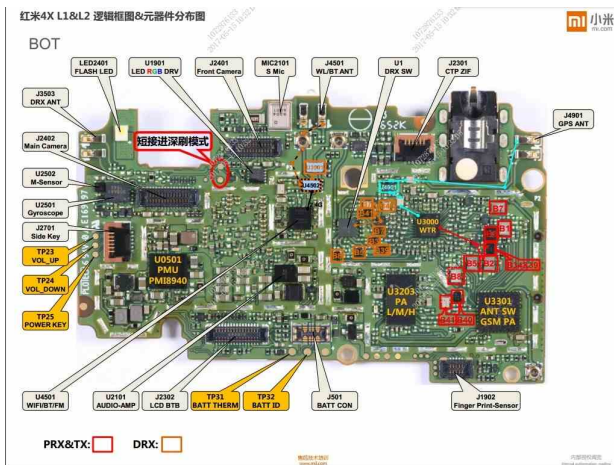
2 Introduction to Motherboard Modules

2.1 Redmi 4X motherboard component distribution diagram

● TOP plane



● BOT surface



2.2 Brief introduction of red rice 4X startup timing and key signal measurement table

Boot timing diagram:

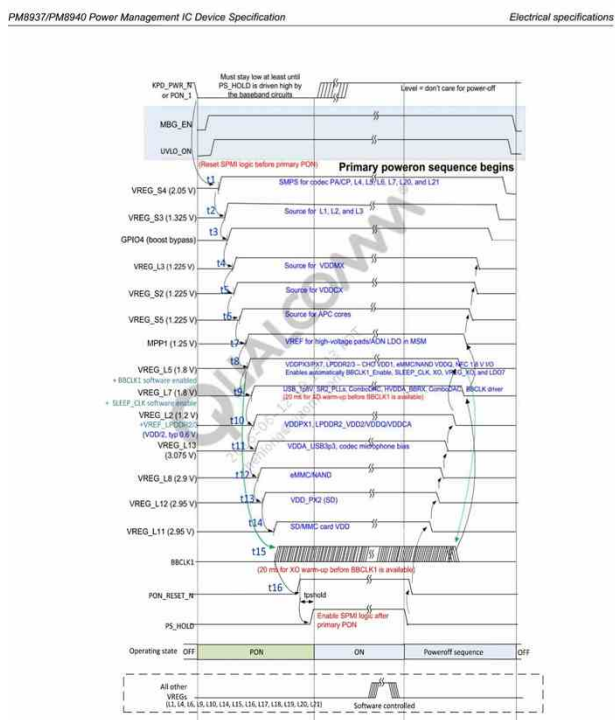


Figure 3-8 PM8937/PM8940 power-on sequence

Figure 3-8 PM8937/R Boot Timing:

Symbol	测量值	测量点	Application
KYPD_PWR_N	1.79V	TP25	PWR_N FROM U0501~
VREG_S4_2P25	2.05V	C1004	Power for High-voltage LDOs
VREG_S3_1P25	1.3V	C810	Power for Low-voltage LDOs
VREG_S2_1P25	1.2V	C903	Power for on-chip memory
VREG_S2_1P25	1.23V	L0802	Power for MSM Core/GRA
VREG_S5_S6_APC_1P25	1.25V	C1545	Power for MSM APC
VDD_PX_BIAS_MPP_1	1.24V	C1510	VDD_PX_BIAS_MPP_1
VREG_L4_1P8	1.70V	R0305	IO/UART
VREG_L2_1P8	1.70V	C9007	Analogs
VREF_LPD0R3	1.2V	C1828	LP0R0AMP1P1
VREF_LPD0R3	0.6V	C1810	LP0R0参考电压
VREG_L3_3P075	3V	C0913	USB and AUDIO
VREG_L2_2P9	2.9V	C0923	摄像头
VREG_L12_PX2_2P95	2.96V	C0930	SEC2(SD卡) 模块供电
VREG_L11_2P95	2.95V	C0925	Micro SD
PEX1K1	1.16V	C0701	系统时钟
PEX0K1	1.79V		

CP1 (U1101) power supply signal and measuring point:

Symbol	测量值	测量点	Application
VREG_L3_1P225	1.2V	C0903	Power for on-chip memory
VREG_S2_1P225	1.23V	L0802	Power for MSM CoreGPA
VREG_S5_S6_AFC_1P225	1.28V	C1545	Power for MSM AFC
VDD_FX_BIAS_MFP_1	1.24V	C1510	VDD_FX_BIAS_MFP_1
VREG_L5_1P8	1.79V	R0905	IOHUB
VREG_L1_1P8	1.79V	C0907	Analog
VREG_L2_1P2	1.2V	C1828	LPDDR4M1P1
VREF_LF0083	0.6V	C1810	ERIO_CH0_CA基准电压
VREG_L13_3P075	3V	C0913	USB and AUDIO
VREG_L12_FX2_2P95	2.96V	C0930	SEC(ISO+) 模拟电源
VREG_L6_1P8	1.8V	C1502	Power for programming the QFPROM; PLL
VREG_S1_1P225	1.2V	L0801	Power for MSM Modem
VREG_L19_1P3	1.3V	C0919	Power for WLAN
VREG_L14_U1H1	1.8V	C2601	Power for U1H1
VREG_L19_U1H2	1.8V	C2602	Power for U1H2

3 Troubleshooting

3.1 Switch-on and shutdown faults

In the process of maintenance without starting up, we should follow the principle of software before hardware, pay attention to observe whether the motherboard components are damaged, breakdown, liquid inflow, etc., and then measure according to the starting time sequence.

3.1.1 Constant current does not start up

Analysis ideas:

1. Software upgrade, troubleshooting software.
2. If the current is still constant after brushing, measure whether the startup timing and CPU working conditions are normal.
3. If the brush machine reports an error, measure whether the working conditions of U1101 and U1801 are normal.

Maintenance Case 1

Fault phenomenon: No startup,
120mA constant current fault cause:
software failure

Maintenance analysis: 120mA constant current does not start, software upgrade fault repair.

Maintenance Case 2

Fault phenomenon: 110mA constant current

Failure cause: U1801, U1101

Maintenance mode: Do not start up when sending for repair, report error when brushing, pass after replacing U1801, start up 120mA constant current, and repair the fault after replacing U1101.

Maintenance Case 3

Fault phenomenon:
120mA constant current
fault cause: U1101

Maintenance mode: 120mA constant current brush machine reports error and repairs the fault after replacing U1101.

Maintenance Case 4

Fault phenomenon: 250mA constant current, no online fault cause:
U1101

Maintenance mode: Do not start up when sending for repair, do not go online when detecting 250mA constant current, and repair the fault after replacing U1101.

Maintenance Case 1

Fault phenomenon: No startup,
110mA constant current fault cause:
U1101

Maintenance analysis: 110mA constant current is detected without starting up for repair, error is reported by brushing machine, and fault is repaired after replacing U1101.

3.1.2 The startup current is not maintained

Analysis ideas:

1. Software upgrade, troubleshooting software.
2. Measure whether the startup timing signal is normal (whether the signal in the startup timing is caused by short circuit).
3. Measure whether the power supply of U1101 is normal.
4. Measure whether the USB signal line is normal (USB_ID; USB_DM; USB_DP).

Maintenance Case 1

Fault phenomenon: 20mA
does not maintain faulty
component: X0701

Maintenance analysis: No start-up for repair, no maintenance for detection of 20mA, no voltage output for bbclk signal found by measurement, invalid after replacement of suspected power supply problem, invalid after replacement considering U1101 problem, judged as clock circuit

Question, consider that the clock source is X0701, and repair the fault after replacement.

Maintenance Case 2

Fault phenomenon: 230mA
does not maintain fault
component: U1101 virtual
welding

Maintenance analysis: No charge is sent for repair, 230mA is not maintained, no problem is found in measurement, and the fault is repaired after welding U1101.

Maintenance Case 3

Fault phenomenon: 20mA
does not maintain faulty
component: U1101

Maintenance analysis: The failure of sending repair is not started, the detection is not maintained for 20mA, no abnormality is found in the measurement, it is not good after welding U1101, and the failure is repaired after replacement.

Maintenance Case 4

Fault phenomenon: 40mA
does not maintain fault
component: U1101 virtual
welding

Maintenance analysis: Do not start up when sending for repair, do not maintain when detecting 40mA, and repair the fault after taking off the shielding frame.

3.1.3 No current for startup

Analysis ideas:

1. Check whether the appearance of J2701 and J501 is damaged. If the interface is normal, measure whether the ground value of J501 is normal

(BAT_ID, BAT_THERM, VBATT).

2. Power up to measure whether the VPH_PWR output is normal and whether the KYPD_PWR_N1 (TP25) 1.8 V voltage is normal. If there is no output, replace U0701.

3. Measure whether the startup timing signal is normal.

Maintenance Case 1

Fault phenomenon: No startup, no startup current Fault component:

U0501

Maintenance analysis: No power-on for repair, no current detected, short circuit of VBAT to ground found by measurement, troubleshooting after taking off U0501, and repairing after replacement.

Maintenance Case 2

Fault phenomenon: No startup, no startup current Fault component:

U0501

Maintenance analysis: Do not start up when sending for repair, detect no current measurement and find BAT_CON_ID infinity, and repair the fault after replacing U0501.

Maintenance Case 3

Fault phenomenon: No startup, no startup current Fault components:

U0501, U0701

Maintenance analysis: No start-up when sending for repair, no current when detecting, short circuit of VPH and VBAT when measuring, fault repair after replacing U0701 and U0501.

3.1.4 Leakage does not start up

Maintenance analysis ideas:

1. First of all, visually check whether there are components rupture, breakdown, liquid corrosion and discoloration in the appearance of the motherboard.

2. Measure whether VBATT and VPH_PWR are short-circuited. If both VBATT and VPH are short-circuited, find out the short-circuited elements in VBATT circuit first, and then find the short-circuited elements in VPH circuit.

3. Measure whether there is short circuit in other power supply lines, and find out the faulty components according to the short circuit signal.

4. Power on to find heating elements.

3. Measure whether the output voltage line of power supply (U0701, U0501) is normal. Maintenance Case 1

Fault Phenomenon:

Leakage 20mA Fault

Component: U0501

Analysis ideas: Do not start up when sending for repair. It is found that the flash lights are always on after power-on, and the leakage is still 20mA after taking off the flash lights. Troubleshooting after replacing U0501.

Maintenance Case 2

Fault phenomenon: leakage 3A

Faulty components: U1101, U0501, U0701, U2101, U3203

Analysis ideas: Do not start up when sending for repair, detect leakage for 3A, find VPH short circuit by measurement, VPH is normal after replacing U0501, U2101 and U0701, find 150mA constant current after starting up, report error by brushing, eliminate faults after replacing U1101, and repair after replacing U3203 with BAND 7 under calibration.

Maintenance Case 3

Fault phenomenon:
leakage 1.5 A Fault
component: U3301

Maintenance analysis idea: Do not start up when sending for repair, detect the leakage of 1.5 A, measure and find that VBATt-RF is short-circuited to ground, and power it up alone and find that U3301 has serious heat, and repair the fault after replacement.

Maintenance Case 4

Fault phenomenon: 150mA leakage

Faulty components: U1101, U0701

Maintenance analysis ideas: Do not read the card when sending for repair, detect the leakage of 150mA, find that VREG_S5_S6_APC_1P225 is short-circuited to ground by measurement, repair the fault after replacing U0701, find that 110mA is not maintained when starting up, suspect that U1101 is burnt out, and repair the fault after replacing.

3.2 Restart type failure

Analysis ideas:

1. Check whether the KYPD_PWR_N1 1.8 V voltage is pulled low.
2. Software upgrade, troubleshooting software.
3. Measure whether I2C to ground value and voltage are normal.
4. Measure whether the clock is normal when U1101 is powered.
5. Measure whether the motherboard has abnormal power supply caused by short circuit.
6. Replace U1101

Maintenance Case 1

Failure phenomenon: restart
failure during startup cause:
software upgrade

Maintenance analysis idea: After cleaning the factory package of NV, repair the fault.

Maintenance Case 2

Fault phenomenon: White rice
restarts faulty component: U1101
during startup

Maintenance analysis ideas: send repair and restart, check and restart, brush the machine invalid, and repair the fault after replacing U1101.

3.3 Crash type fault

Analysis ideas:

1. Software upgrade, troubleshooting software.
2. Measure whether BAT_ID and BAT_THERM are normal.
3. Measure whether the startup timing is normal.
4. Measure whether the power supply of U1101 is normal.
5. Measure whether there is abnormal power supply caused by short circuit on the motherboard.
6. Replace U1101.

Maintenance Case 1

Failure phenomenon:

White rice fixing

screen failure cause:

software upgrade

Maintenance analysis idea: troubleshooting after starting the white rice fixed screen brush factory package.

Maintenance Case 2

Failure phenomenon: Crash

during startup cause: U1101

virtual welding

Maintenance analysis ideas: send repair crash, crash in the process of detection startup, invalid brush machine, and repair the fault after welding U1101.

Maintenance Case 3

Fault Phenomenon:

Crash while taking

pictures Fault

Component: U0701

Maintenance analysis idea: send repair photo crash, no abnormality is found in measurement, CPU replacement is not good, the current drops to 0V instantly when observing photo, and the power supply problem is considered, and the fault is repaired after replacing U0701.

Maintenance Case 4

Failure Phenomenon: Crash

during startup Failure

Component: U1101

Maintenance analysis ideas: send repair without starting up and charging, detect startup crash, brush invalid, and repair the fault after replacing U1101.

Maintenance Case 4

Fault Phenomenon: Crash

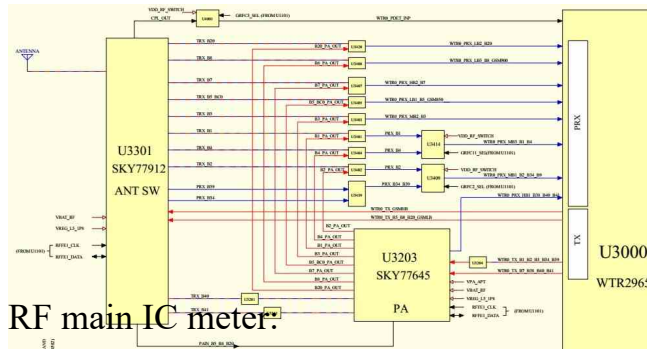
during startup Fault

Component: U0501

Maintenance analysis ideas: send repair without starting up, detect startup crash, brush invalid, show low battery power, measure THEMR without problem, and repair the fault after replacing U0501.

3.4 Signal related function failure

Red rice 4X RF circuit block diagram:



IC	Symbol	测量值	测量点
U3000	WTR XO CLK	19.2	R3000
	VREG L1 1P0	1V	C0901
	VREG L4 1P8	1.8V	R3005
U3203	VPA APT	4V	R3210
	VREG L5 1P8	1.8V	R0905
	VBAT RF	4V	R3302
U3301	VBAT RF	4V	R3302
	VREG L5 1P8	1.8V	R0905
	VDD RF SWITCH	2.7V	R3101
U1	VREG L5 1P8	1.8V	R0905

1. Insert SIM card to ensure normal identification, and eliminate faults that do not recognize SIM card.
2. Software upgrade, troubleshooting software.
3. RF calibration, check the specific test items through the test report. However, measure the RF path according to the corresponding system and principle block diagram to find the fault point.
4. Measure whether the power supply of RF circuit is normal and whether the RFFE (1/2) CLK/DAT signal is normal.
5. If the RF calibration is normal and there is still no signal, check the SIM card circuit and whether the U3101 to J3101 circuit is normal.

Maintenance Case 1

Fault Phenomenon:

Signal Weak Fault

Component: U3301

Analysis ideas: The signal is weak, the RF components of the motherboard are burned and the appearance of the motherboard is deformed.

Maintenance Case 2

Fault Phenomenon:

No Signal Fault

Component: U3203

Analysis ideas: no signal, calibrate and report error LTE Band39, open the RF shielding case and find that U3203 has burning marks, measure the low value of VBAT_RF to ground, and repair the fault after replacing U3203.

Maintenance Case 3

Fault phenomenon: no signal

Faulty component: U3203

Analysis ideas: No signal, calibrate error WCDMA BAND1, measure short circuit of VPA_APT, remove short circuit of U3203, repair fault after replacement.

Maintenance Case 4

Fault phenomenon: no signal

Faulty component: U1101 (virtual welding)

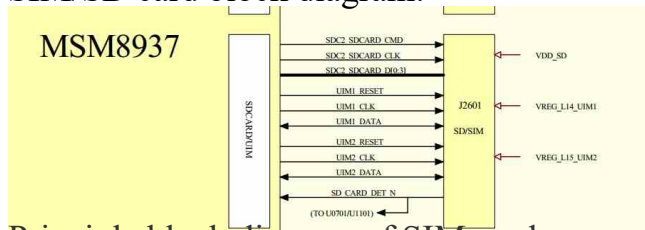
Analysis ideas: No signal, calibration error WCDMA BAND1, welding U1101 after the fault repair.

3.5 SD/SIM card related function failure

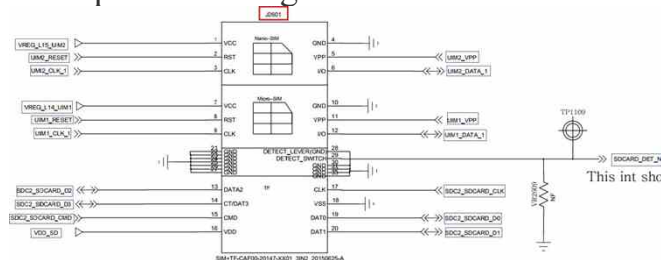
Instructions for using dual cards:

Micro + Nano/TF can be selected, SIM card of any standard can be inserted into card slot 1, and TF (Micro SD) card can be inserted into card slot 2.

SIM/SD card block diagram:



Principle block diagram of SIM card:



Maintenance analysis ideas:

1. First, check whether the mobile phone has baseband information. If there is no baseband information, it will be repaired according to no baseband failure. If the baseband information is normal, it will be a functional failure of SIM card.
2. Check whether the J2601 contact is deformed or damaged. If the appearance is normal, press the J2601 position of the motherboard to see if the card is read

The card slot problem can lead to poor contact of the card.

3. Measure whether the SD/SIM card holder ground value is normal and whether the test point to the card holder is disconnected (to eliminate false soldering, measure whether the path from the test point to the card holder pad is normal).

4. Start up and measure whether the power supply, clock, reset and data voltage jump of SIM card are normal. If the SD card cannot be identified, measure whether the power supply of the SD card is normal.

5. If the above signals are normal, replace U1101 and U1801.

Maintenance Case 1

Fault Phenomenon: Not aware of SIM card Fault
Component: J2601

Maintenance analysis: The card is not recognized when it is sent for repair, and the SIM card slot is found to be damaged, and the fault is repaired after replacement.

3.6 Baseband unknown

Maintenance analysis ideas:

1. Deep brush factory software (erase NV), do RF calibration and troubleshoot software.
2. Measure whether RFFE1_DATA and RFFE1_CLK are abnormal.
3. Measure whether the power supply of U1101 is normal.

Maintenance example

Failure phenomenon: invalid IMEI, no signal, no card recognition Failure cause: software failure

Maintenance analysis: Qualcomm software BUG leads to baseband loss.

3.7 Charging related function failure

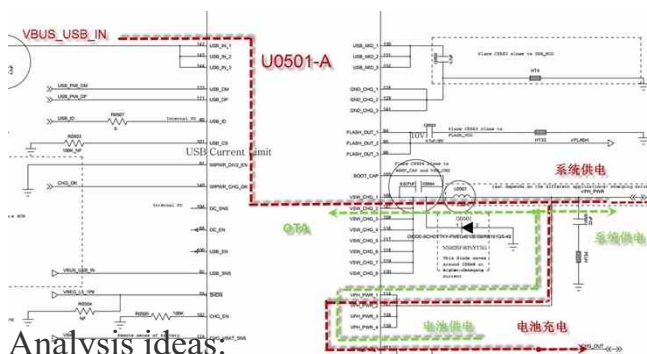
SW charging process is divided into five states: battery activation, precharge, constant current charging, constant voltage charging and recharging.

In the battery activation stage, the battery voltage is less than 2V, and the charging current at this time is 30mA;; Once the battery voltage reaches 3V, precharge begins, and the current is generally set to 500mA. Charge to the startup voltage point set by the system

(3.4 V), that is, it enters the constant current charging stage. When the constant current charging reaches the cut-off voltage set by the system of 4.34 V, it starts to enter the constant voltage charging state. At this time, the charging current will slowly drop to the cut-off current set by the system of 97mA, and the system stops charging, then enters the discharge mode, and then enters the recharging state when the discharge is less than the cut-off voltage of 120mV.

During the whole process, the battery activation and precharge stages are very short, which is difficult to measure, and most of the time is in CC stage.

Charging schematic diagram:



Analysis ideas:

1. Software upgrades troubleshoot software.
2. Check whether the appearance of J1901 is normal, and measure whether the four signals of J1901 are normal to ground (VBUS_USB_IN, USB_HS_D_P, USB_HS_D_M, USB_ID).

- 3.Measure whether (VBAT, BAT_CON_ID, BAT_CON_THERM) of J501 is normal.
4. With charging state, no charging current or low charging current (current is generally tens of milliamperes), replace U0501
- 5.If the above signals are normal, replace U1101 and U1801.

Maintenance Case 1

Fault Phenomenon:

No Charge Faulty

Component: U1903

Maintenance analysis ideas: no charging for repair, no charging for detection, short circuit of VBUS_USB_IN_1 found by measurement, burning of U1903 found by checking the motherboard, and repair the fault after replacement.

Maintenance Case 2

Fault Phenomenon:

No Charge Faulty

Component: U1101

Maintenance analysis ideas: send repair without charging, test without charging, measure found ID to ground value 190, good board 500, replacement U0501 invalid judgment U1101 problem.

Maintenance Case 3

Fault Phenomenon:

No Charge Faulty

Component: U0501

Maintenance analysis ideas: no charging for repair, no charging for detection, no problem found in measurement, and fault repair after replacing U0501.

Maintenance case

Fault Phenomenon:

No Charge Faulty

Component: U1101

Maintenance analysis ideas: Do not charge in the startup state, brush the machine without knowing the mouth, measure DM, DP and USB ID normally, and repair the fault after replacing U1101.

3.7 Display related function failures

J2302 is the connection interface of display screen, backlight driver and display driver are generated by U0501, and VREG_L6_1P8 supplies power for LCD I/O.

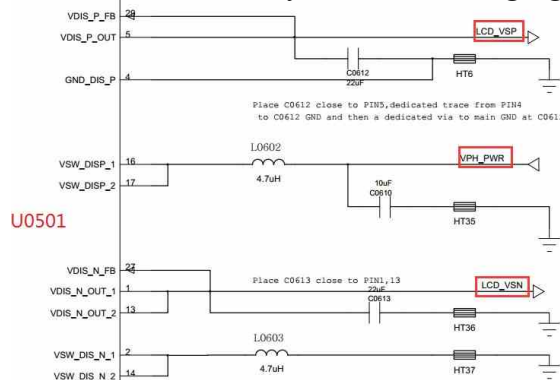
J2302 Meter: MIPI

Symbol	测量值	测量点	Application
VREG_L6_1P8	1.8V	C2304	I/O供电From U0501
LCD_RST_N	1.8V	R2307	RST From U1101
VSP	+5V	C2350	LCD显示驱动From U0501
VSN	-5V	C2351	
LEDA	15.5V~18.5V	C2355	LCD背光电压From U0501

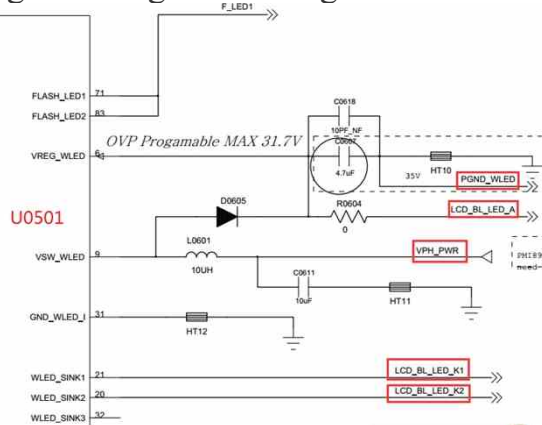
MIPI总线			
Symbol	测量值	测量点	Application
MIPI_DSIO_LANE_N[0:3]	348	J2302接口:2/5/11/14	J2302通过滤波元件 EM[2301、2302、2304、 2305]连接到CPU
MIPI_DSIO_LANE_P[0:3]		J2303接口:3/6/12/15	
MIPI_DSIO_CLK_N		J2304接口:8	J2302通过滤波元件 EM2303连接到CPU
MIPI_DSIO_CLK_P		J2305接口:9	

U0501 Display Drive Circuit Diagram:

VSP, VSN $\pm 5V$ standby without voltage generation.



Backlight driving circuit diagram:



PWM OUT waveform diagram:



Maintenance analysis ideas:

1. Visually check whether the appearance of J2302 is damaged or falsely welded.
2. Measure whether J2302 is normal to ground with multimeter diode gear.
3. If the ground value is normal, measure whether VSP, VSN and VREG_L6_1P8 have output.
4. Measure whether LCD_BL_LED_A voltage is normal. If abnormal, replace U0501 and boost components.
5. Replace U1101 and U1801.

Maintenance Case 1

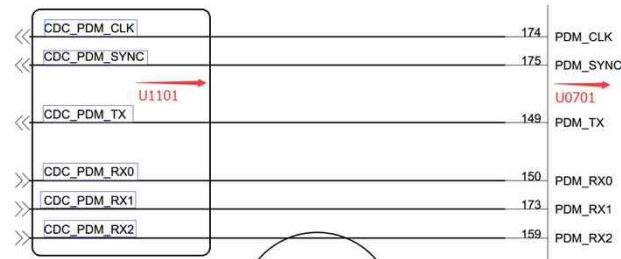
Fault Phenomenon:
Black Screen Fault
Component: U0501
Maintenance analysis ideas: send to repair black screen to detect black screen, measure VSN 3V and VSP 0V, and repair the fault after replacing U0501.

Maintenance Case 2
Fault Phenomenon:
No Display Fault
Component: L0601
Maintenance analysis idea: There is a problem with the black screen detection boost circuit for repair, and it is invalid to replace U0501. Find the circuit and find it is L0601.

Maintenance Case 3
Fault Phenomenon:
No Display Fault
Component: C0607
Maintenance analysis ideas: Black screen, measure C0607 short circuit, observe that this component has burnt traces, and repair the fault after replacement.

3.8 Audio related function failure

The audio signal between U1101 and U0701 is transmitted by PDM bus: CDC_PDM_TX is used for MIC signal
CDC_PDM_RX0 for earphone signal
CDC_PDM_RX1 for speaker signal
CDC_PDM_RX2 for earphone
earphone signal



3.8.1 Loudspeaker failure

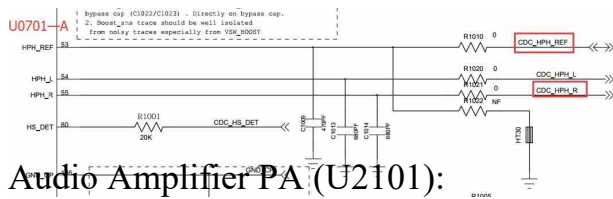
PM8937 power IC is selected, codec part is integrated in PM8937, with independent SPK PA (U2101).

The main power supply measurements for the codec section are as follows:

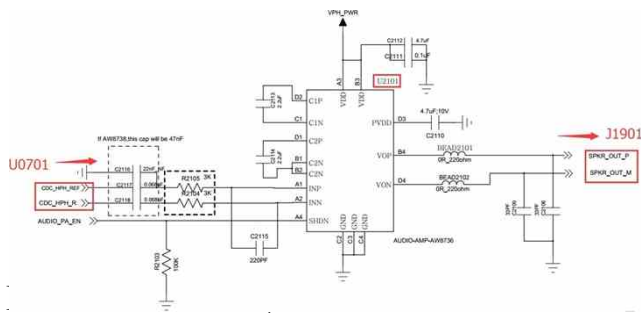
Symbol	测量值	测量点	Applicatio.
VREG_L5_1P8	1.8V	C1005	CDC的IO供电U0701产生
VREG_S4_2P05	2.05V	C1020	扬声器音频放大器供电
VPH-PWR	4V	C1024	音频供电驱动

The audio signal from PMU CODEC (PM8937) has two signals CDC_HPH_R and CDC_HPH_REF as the audio input of PA (U2101), which are amplified by U2101 and output SPKR_OUT_P and SPKR_OUT_M to J1901, and then play the sound in the speaker.

PM8937 Audio Output:



Audio Amplifier PA (U2101):



1. Software upgrade, troubleshooting software.
2. Measure whether the CODEC power supply is normal (considering whether the charging pump circuit is normal).
3. Open the audio file, and measure whether the waveforms of CDC HPH R and CDC HPH REF signals output by U0701 are normal with oscilloscope according to the channel (compare the waveforms at both ends of the capacitor at measuring points C2118/C2117 to judge whether the capacitor is damaged, and analyze in Step 5 if the signal is normal).
4. Take off U0701 and use multimeter diode to measure whether PDM bus is normal to ground.
5. Measuring U2101 outputs SPKR_OUT_P and SPKR_OUT_M signals (measuring points C2109/C2108). If there is any abnormality, detect the working conditions and body of U2101.

Maintenance Case 1

Fault phenomenon: speaker silent and
standby crash Fault component:

U0701

Maintenance analysis ideas: Send the loudspeaker for repair, brush the machine is invalid,
and repair the fault after replacing U0701.

Maintenance Case 2

Fault Phenomenon:
Speaker Silent Fault
Component: C1016

Maintenance analysis ideas: The loudspeaker for repair is silent, the loudspeaker for
detection is silent and has noise, C1016 is found to be burned by measurement, and the
pad is found to be burned after taking it off.

Maintenance Case 3

Fault Phenomenon:
Speaker Silent Fault
Component: U2101

Maintenance analysis ideas: There is noise in the repair recording, the actual detection
speaker is silent, the replacement of U0701 is invalid, and the fault is repaired after the
replacement of U2101.

3.8.2 Don't send a message:

MIC IN3 P



- ### 3.8.3 Handset failure:

[illegible]

Maintenance analysis ideas:

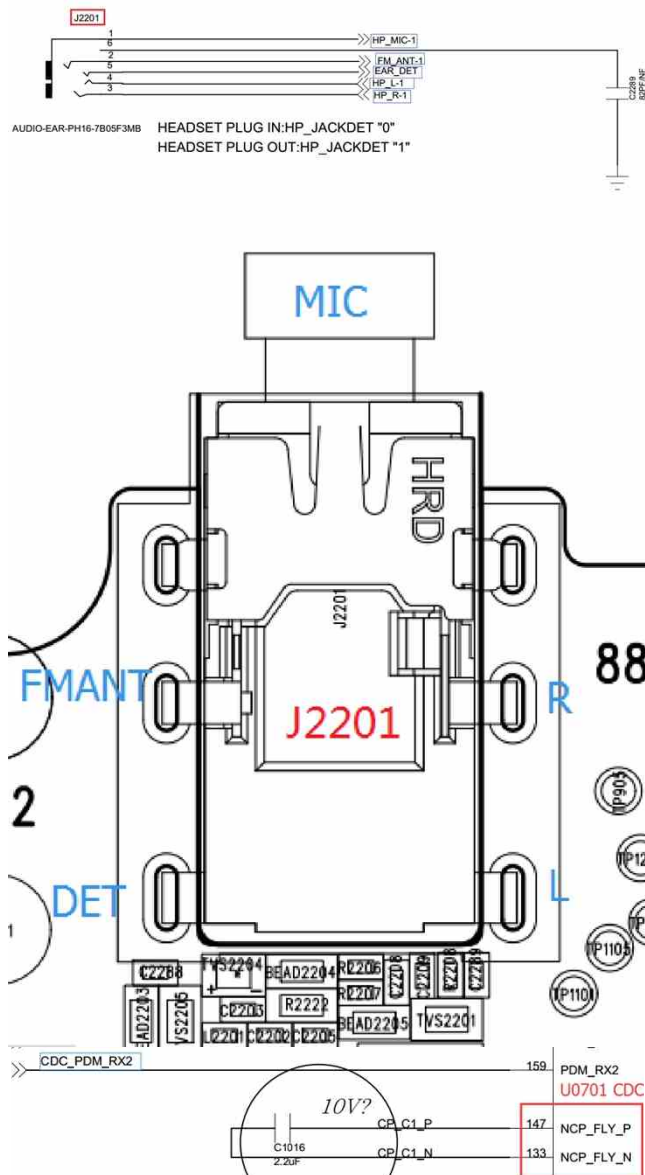
- ### 3.8.4 Headphone failure:

Maintenance analysis ideas:

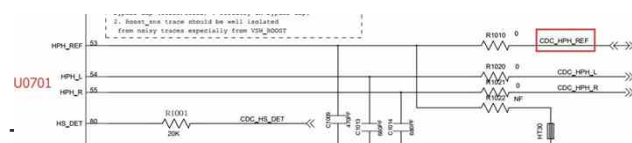
1. Check whether the earphone interface contacts are deformed and whether there are foreign bodies in the earphone interface.
2. Software upgrade, troubleshooting software.
3. Measure whether the earphone interface to the ground value is normal, do not insert the earphone when starting up, and measure whether the DET detection pipe angle has

1.8 V Voltage

4. Repair the headphone channel according to the principle block diagram, and replace U1100 normally. Headphone interface pin:



6. Turn on the audio plug-in headset test to see if CDC has audio output. If the output signal is abnormal, focus on detecting whether the CDC HPH REF to the earphone interface FM ANT in the figure is open.



the output audio signal is normal, replace U2201.

Maintenance example 1

Failure Phenomenon:

Headphone Mode

Failure Cause:

TVS2201

Maintenance analysis: send earphone repair mode, test earphone mode, measure EAR_DET to ground resistance value 220, good board infinity, replace TVS2201 after the fault repair.

Maintenance example 2

Failure phenomenon:

earphone microphone silent

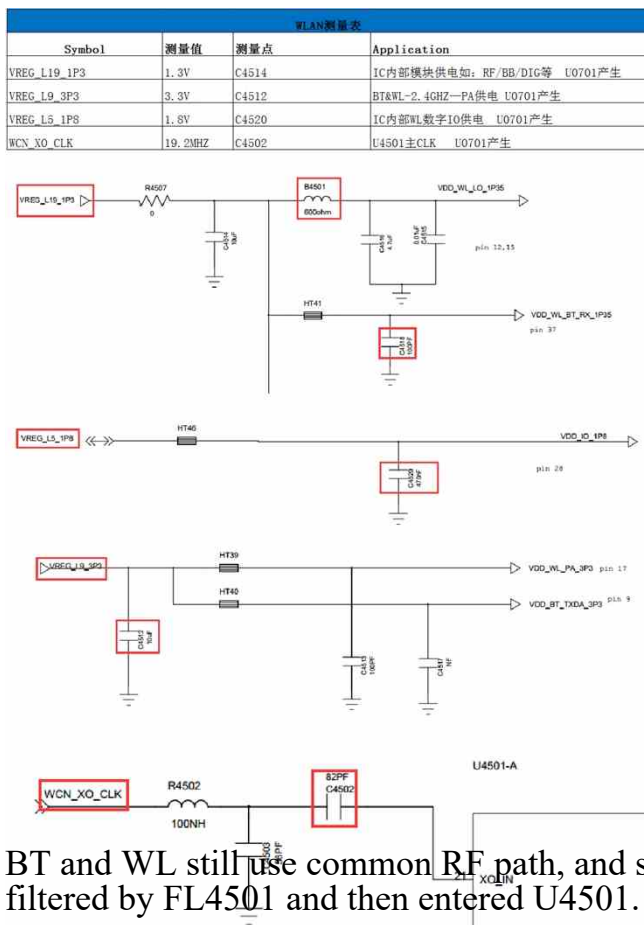
failure cause: U0701

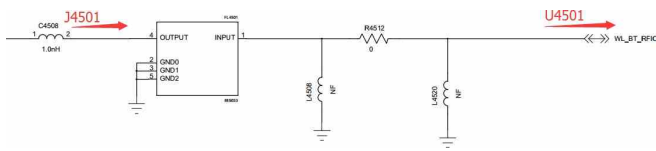
Maintenance analysis: The earphone microphone is sometimes silent for repair, and the earphone microphone is silent for detection. No problem is found in measurement, and the fault is repaired after replacing U0701.

3.9 WIFI/BT/FM related function failure

Principle: WIFI/BT/FM is integrated in U4501, CPU realizes communication and control to U4501 through bus, and U0701 provides power supply to U4501. U4501 is not equipped with a clock crystal oscillator, and the power output WCN_XO_CLK (19.2 MHz) is used as the main clock of U4501.

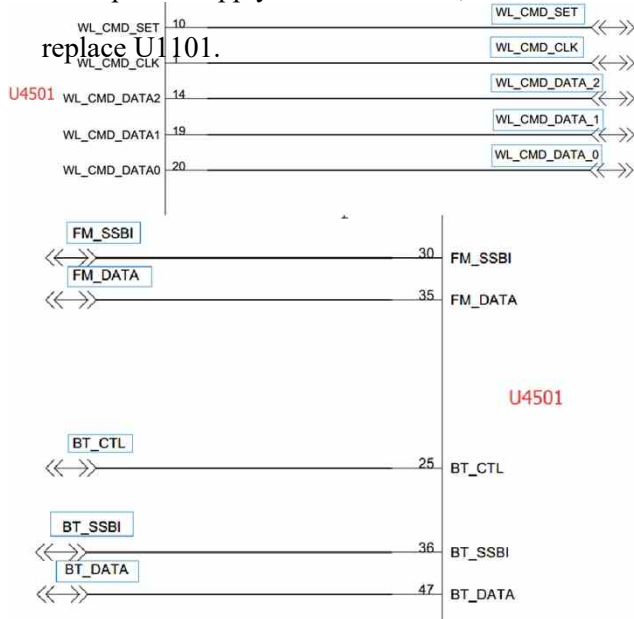
U4501 Power Supply and Clock:





Maintenance analysis ideas:

1. Brush the machine to eliminate software problems.
2. If the WIFI signal is weak, measure whether the receiving/transmitting path is abnormal and replace FL4501 normally.
3. Check whether the power supply and clock are normal according to U4501 meter.
4. If the power supply clock is normal, measure the communication line between U4501 and U1101 and



5. All the above signals are normal. Replace U4501.
6. Finally, try to replace U1101.

Maintenance example 1

Fault phenomenon: wifi
no signal fault
component: software
fault

Maintenance analysis: No signal can be found in the wifi for repair, no signal can be found in the wifi for inspection, and the fault can be repaired after brushing the machine.

Maintenance example 2

Fault phenomenon: Bluetooth and
WiFi can't find the signal. Fault
component: U4501

Maintenance analysis: Bluetooth and WiFi can't be found, and the fault can be repaired after replacing U4501.

3.10 Camera related function failure

Camera failures are mainly divided into primary and secondary camera failures, primary

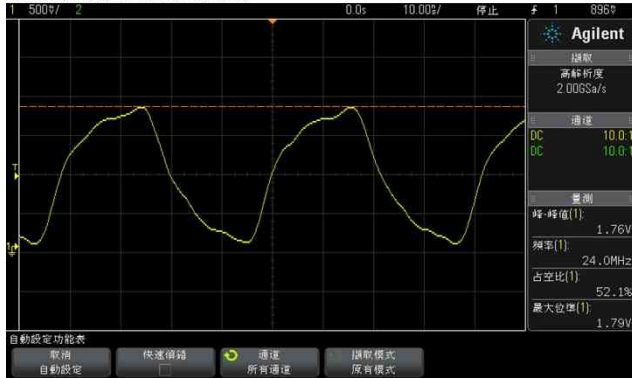
camera failures and secondary camera failures. According to the measurement table, when the common signal of primary and secondary cameras goes wrong, there will be a fault that the front and back can't take pictures, and of course, other signals can't be damaged at the same time.

Camera measurement signal table:

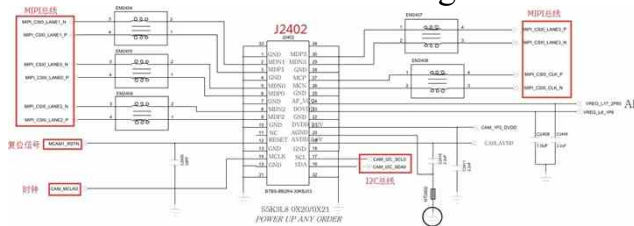
元器件	Symbol	测量值	测量点	Application
J2402/J2401	CAM_1P2_DVDD	1.2V	C2411	内部逻辑电路供电
	VREG_L6_1P8	1.8V	C2408	I/O供电
	CAM_AVDD	2.8V	C2410	基础供电
	CAM_I2C_SDA0	1.8V	R1306	I2C总线
	CAM_I2C_SCL0	1.8V	R1305	
J2402	VREG_L17_2P85	2.85V	C2409	AF（自动对焦）供电
	MCAM1_RSTN	1.8V	C2450	MCAM复位信号
	CAM_MCLK0	1.76V	R1309	MCAM时钟
J2401	SCAM_RST_N	1.8V	C2451	SCAM复位信号
	CAM_MCLK2	1.76V	R1318	SCAM时钟
	SCAM_PWDN	1.8V	C2452	相机关闭信号

CAM0_MCLK/CAM1_MCLK:

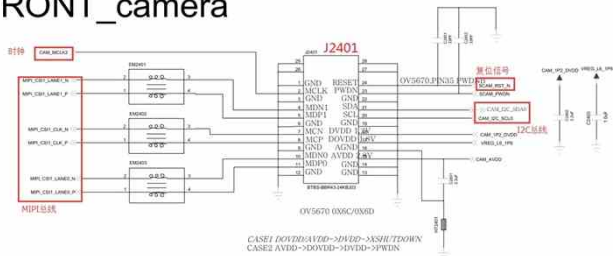
090-X-2022A M/53160144 Wed Feb 03 17:23:15 2016



Main camera J2402 circuit diagram:



Circuit diagram of sub-camera J2401: FRONT_camera



Maintenance analysis ideas:

1. Software upgrade, troubleshooting software problems.
2. Carefully observe whether each pin of J2402 and J2401 has virtual welding or depression. If normal, measure whether the ground value of the interface is normal.
3. Measure whether the camera power supply, clock and reset signal output are normal.

4. If the above signals are normal, replace U1101. Maintenance Case 1

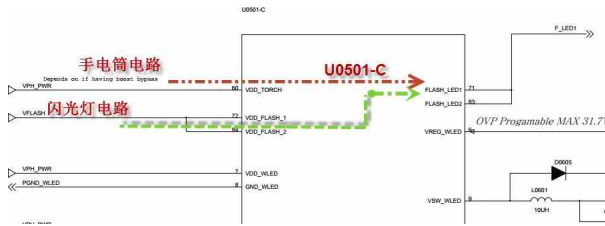
Fault Phenomenon: Post-camera does not photograph Fault Component: U1101

Virtual Welding

Maintenance analysis ideas: the camera for repair cannot be opened, the rear camera for detection does not shoot, no problems are found in measurement, and the fault is repaired after welding U1101.

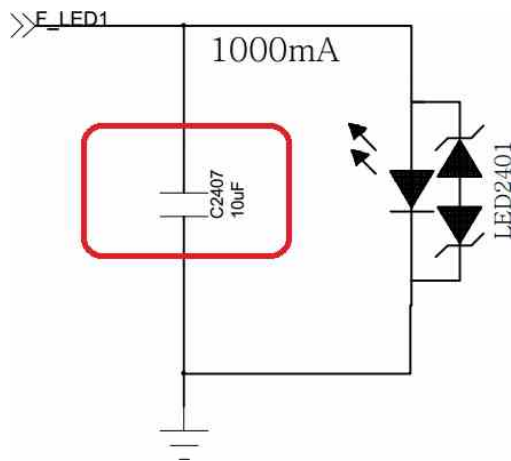
3. 1 Flash related function failure

Flash circuit:



Maintenance analysis ideas:

1. Software upgrade, troubleshooting software problems.
2. Measure whether the impedance and voltage of VFLASH to ground are normal, and replace U0501 if it is abnormal.
3. If the flash lamp is normal and the flashlight cannot be turned on, please refer to the maintenance method of Redmi 3S products and remove C2407.



4. Replace LED2401.

Maintenance Case 1

Fault phenomenon: Flash lamp is not bright
Fault component: LED2401

Maintenance analysis idea: The flash lamp for repair is not bright, and the fault is repaired after replacing LED2401.

3.12 Touch screen function failure

Touch screen circuit signals are: power supply, INT, RST, I2C, these signals can be directly measured without contact screen voltage.

TP meter:

Symbol	测量值	测量点	Application
TP_RST_N	1.8V	TP901	TP复位信号 From U1101
TP_INT_N	1.8V	TP902	TP中断信号 From U1101
VREG_L10_2P8	2.8V	TP903	TP供电
TP_I2C_SDA	1.8V	TP904	I2C总线
TP_I2C_SCL	1.8V	TP905	

Maintenance example 1

Failure phenomenon:

touch screen failure

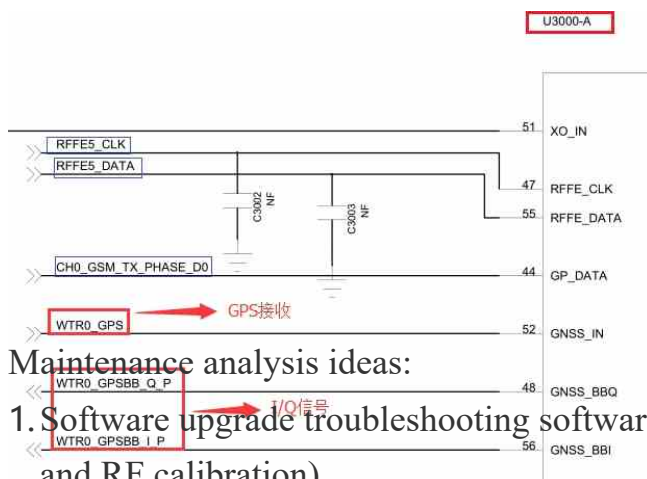
Failure cause: C1902

Maintenance analysis: the touch screen for repair failed, the detection failed, the short circuit of VREG_L10_2P8 was found by measurement, and the problem was found to be C1902 after being removed in turn.

3.13 GPS related function failure

GPS is integrated in U3000 (intermediate frequency),

GPS receiving path: GPS-IQ signal:



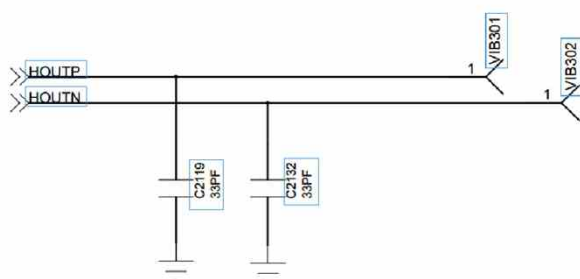
Maintenance analysis ideas:

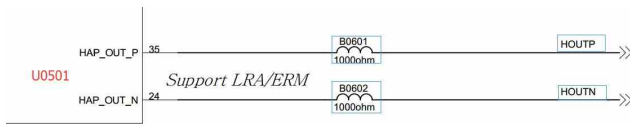
1. Software upgrade → I/Q信号 troubleshooting software (including factory software for erasing NV and RF calibration).
2. Measure whether the GPS receiving channel is normal.
3. Measure whether the working conditions of U4901 are normal.
4. Take off U3000 to measure whether IQ signal is normal, and replace U3000 normally.
5. If the above signal is normal, replace U1101.

3.14 Vibration related function failure

When the working voltage of the vibrator is 2.7 V, the vibration state is the best. Vibrator working signal:

Haptics			
48	VDD_HAP	PI	Power supply for Haptics circuits
36	GND_HAP	GNDP	Haptics power ground
35	HAP_OUT_P	AO	Haptics H-bridge driver output positive
24	HAP_OUT_N	AO	Haptics H-bridge driver output negative
11, 12	HAP_PWM_IN	DI	PWM input for haptic control





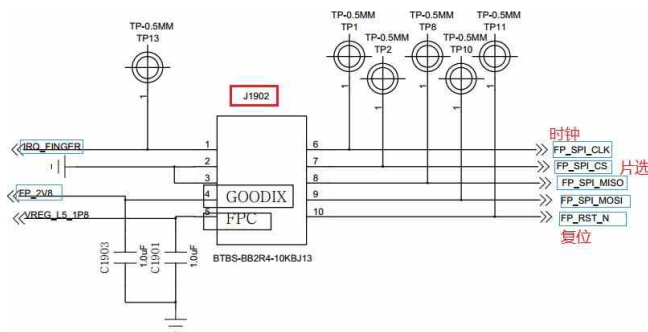
Maintenance analysis ideas:

1. Software upgrade, troubleshooting software.
2. Measure whether the value of VIB301 and VIB302 is normal, and replace U0501 if it is abnormal.
3. Measure whether the control signal GP_PDM_A0 is normal, and replace U1101 if it is abnormal.

3.14 Fingerprint related function failure

Fingerprint circuit signals are: power supply, CS, RST, these signals are not connected with fingerprints can also directly measure its voltage.

Symbol	测量值	测量点	Application
FP_2V8	2.8V	C1903	基础供电
VREG_L5_1P8	1.8V	C1901	I/O供电
FP_SPI_CS	1.8V	TP2	片选信号
FP_RST_N	1.8V	TP11	复位信号



Maintenance analysis ideas:

1. Software upgrades troubleshoot software.
2. Measure whether the resistance values of TP1, TP2, TP8, TP10, TP11 and TP13 to ground are normal, and whether the voltages of TP2, TP11, C1901 and C1903 are normal.
3. Check whether U1902 is normal.
4. Replace U1101.